REDKROSS RESEARCH FOUNDATION

Complete Architecture Blueprint

From mission and governance to education, cyber labs, research, innovation and public impact

LEARN	PRACTISE	INVESTIGATE	CREATE IMPACT
Academy, curriculum, assessments	Isolated labs, CVEs, CTFs	Research questions, methods, evidence	Tools, publications, community outcomes

Architecture intent

RedKross should operate as one connected research and learning ecosystem. People learn through the Academy, practise in a controlled Cyber Range, investigate questions through Research, turn validated findings into useful technology through Innovation, and return knowledge and benefit to the community. Governance, safety and quality assurance apply across every domain.

Status: working architecture and implementation reference. Proposed controls, roles, budgets, service levels and technical designs require formal approval before being treated as policy.

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How to use this blueprint

This document sets out a target operating model. It combines organisational structure, governance, programme delivery and technical architecture. It does not imply that every component is already built or staffed. Use it to prioritise decisions, assign ownership, guide student projects and plan implementation.

Contents

- 1. Executive overview and outcomes
- 2. Mission and operating principles
- 3. Organisational architecture
- 4. Governance, risk and decision rights
- 5. People and programme lifecycle
- 6. Academy architecture
- 7. Cyber Range, CVE labs and CTF
- 8. Research lifecycle and reproducibility
- 9. Innovation and engineering
- 10. Digital platform and data architecture
- 11. Security, privacy and safeguarding
- 12. Quality assurance and release
- 13. Operating cadence and end-to-end workflows
- 14. Deliverables and documentation
- 15. Implementation roadmap
- 16. Metrics, risks and open decisions

Stable architecture rules

- **One connected system:** all domains share ownership, governance and traceable work records.
- **Control plane separated from lab plane:** vulnerable labs must not be directly exposed to production systems.
- **Risk-based approvals:** approvals scale with impact, exposure and sensitivity.
- **Evidence before claims:** findings, QA and impact claims require traceable evidence.
- **Independent QA:** higher-risk or public-facing work is reviewed by someone other than its author.
- **Safe by default:** use public sources, synthetic data and authorised environments.

1. Executive overview and outcomes

RedKross is not only a website, internship programme, course catalogue or cyber lab. It is an ecosystem that develops people and produces useful, evidence-backed outcomes. The organisation should operate through a repeatable loop: govern, onboard, learn, practise, investigate, build, assure, share, measure and improve.

Stage	Purpose	Output
Govern	Set priorities, boundaries, ethics, risk and accountability.	Approved priorities, policies, owners and decisions.
Onboard	Place people in defined programmes with clear scope and supervision.	Role, access, learning plan and project assignment.
Learn	Teach structured concepts through the Academy.	Progress, exercise and assessment evidence.
Practise	Apply skills in isolated labs and controlled CTFs.	Lab results, validation, feedback and reset.
Investigate	Test questions using reproducible methods.	Methods, data, evidence, findings and limitations.
Build	Turn validated findings into tools or prototypes.	Reviewed code, tests, documentation and prototype.
Assure	Check safety, quality, privacy and readiness.	QA result, approval and release record.
Share	Publish responsibly and return knowledge to the community.	Learning materials, publications, open-source outputs.
Improve	Use feedback, incidents and outcome measures to adapt.	Updated curricula, controls and priorities.

Success should be measured by more than activity counts. Look for improved competency, reproducible research, safe lab operation, useful tools, quality public resources, responsible disclosure, community reuse and lessons learned.

2. Mission and operating principles

The mission should be framed around public benefit: improve cybersecurity understanding and capability, enable responsible research, teach practical skills safely, and convert validated knowledge into useful resources and technology.

Principle	Operational meaning
Public benefit	Prioritise work that improves safety, skills, knowledge or community capability.
Ethics and legality	Define permission, scope, consent, rights and disclosure before work begins.
Evidence-led work	Keep source citations, methods, test results, decisions and limitations.
Learning by doing	Connect lessons with safe practice, feedback and reflection.
Least privilege	Give each person only the access required for their role.
Privacy by design	Minimise data, restrict access, redact evidence and define retention.
Accessibility	Make materials and services usable by their intended audience.
Reproducibility	Version code, lab definitions, methods and instructions.
Continuous improvement	Use defects, incidents, feedback and outcome measurements.

3. Organisational architecture

The organisation has six connected domains supported by two cross-cutting foundations. The domains describe the work; the foundations provide rules, evidence and infrastructure used by every domain.

Domain	Responsibility	Typical outputs
Governance & Trust	Ethics, legal/compliance, safeguarding, risk, audit and approvals.	Policies, risk records, approvals, incident actions.
People & Programmes	Students, interns, fellows, mentors, researchers and contributors.	Onboarding, assignments, supervision, completion records.
Academy	Learning paths, courses, modules, lessons, exercises and assessments.	Curriculum, rubrics, progress and certificates.
Cyber Range	Isolated labs, CVE exercises, CTFs, VM/container/hybrid environments.	Lab definitions, images, flags, validation, reset/destroy.
Research	Research areas, projects, experiments, datasets, evidence and review.	Protocols, reproducible artifacts, reports, publications.
Innovation	Convert validated knowledge into useful software and prototypes.	Requirements, code, tests, releases, maintenance.
Community & Impact	Awareness, partnerships, open source and public education.	Guides, events, contributions and impact reports.

Cross-cutting foundations

- **Digital control plane:** public website, private dashboards, identity/access, project records, learning progress, approvals, reporting and notifications.
- **Assurance and evidence:** QA, audit logs, source references, test evidence, release records, retention and incident learning.

Four-part mental model

Area	Plain-language purpose
Academy = Learn	Build understanding and foundational competence.
Cyber Range = Practise	Apply skills in controlled, disposable environments.
Research = Investigate	Test questions and produce evidence-backed knowledge.
Innovation = Make useful	Turn validated knowledge into tools, systems or improvements.

CTF events, CVE labs, forensics, red/blue exercises and cloud labs are formats within this model, not separate organisational silos.

4. Governance, risk and decision rights

Governance establishes which work may proceed, who approves it, what evidence is required and who is accountable. It applies to technical and non-technical work.

Function	Responsibilities
Leadership/programme owner	Set priorities, resources, outcomes and accountable owners.
Ethics/safeguarding	Assess participant welfare, consent, sensitive topics and supervision.
Legal/compliance	Check authority, rights, contracts, licences, privacy and disclosure duties.
Risk owner	Assess threats, impact, likelihood, mitigation and residual risk.
Security owner	Review access, architecture, secrets, isolation, dependencies and incident readiness.
Quality auditor	Independently check deliverables against acceptance criteria.
Project lead	Manage scope, tasks, dependencies, milestones and delivery.
Release approver	Decide whether a publication, lab, tool or deployment is ready.

Risk-based gates

1. Define purpose, scope, owner, audience, expected output and exclusions.

2. Classify sensitivity, system criticality, possible harm and exposure.
3. Authorise access, testing, data use and publication in writing where required.
4. Build or investigate only within the approved scope and environment.
5. Validate correctness, reproducibility, safety, privacy and operational behaviour.
6. Obtain required independent QA and security/ethics/legal review.
7. Approve, request changes, restrict distribution, defer or reject.
8. Monitor, retain appropriate evidence and close corrective actions.

A previous working rule was 10% QA sampling for R0 work and mandatory QA for R1 and above, with one designated QUALITY_AUDITOR. Confirm the active risk definitions, sampling method and approval authorities before operationalising this rule.

5. People and programme lifecycle

Each student, intern, fellow, mentor or contributor should enter through a defined programme and receive a role, supervisor, scope, access level, expected deliverables, cadence and completion criteria. Group chats are useful for coordination but should not be the only permanent record of requirements or approvals.

Lifecycle

1. Recruit or receive an application and communicate programme purpose, terms and expectations.
2. Verify only the information needed for the programme.
3. Assign cohort, supervisor, role, workstream and onboarding checklist.
4. Provide acceptable-use, privacy, safeguarding, security and escalation guidance.
5. Grant least-privilege access to the tools required for the task.
6. Agree learning plan, project brief, milestones, deliverables and review schedule.
7. Track progress through evidence and status updates.
8. Review work, provide feedback and address deficiencies.
9. Offboard: revoke access, transfer work, record completion and apply retention rules.

Roles

Role	Responsibility
Student/intern/fellow	Complete assigned work, document evidence and raise blockers.
Mentor/supervisor	Coach, clarify expectations, review progress and escalate risks.
Project lead	Own project plan, tasks, dependencies and final delivery.
Quality auditor	Independently check acceptance criteria; do not approve own work.
Programme coordinator	Maintain cohort records, onboarding, cadence and communication.
Administrator	Manage accounts, access and records under approved procedures.

6. Academy architecture

The Academy turns knowledge into structured learning. A course is not complete merely because lessons exist: objectives, exercises, feedback and evidence of learning are required.

Level	Purpose and required elements
Learning path	Target role/competency, entry requirements, outcomes and progression.
Course	Related modules, course outcomes, prerequisites and assessment approach.
Module	Objectives, concepts, examples, readings and practical activities.
Lesson	Focused explanation, examples, references and accessibility checks.
Exercise/lab	Instructions, safe environment, expected outcome, hints and reset.
Assessment	Questions/tasks, answer key or rubric, scoring and feedback.
Certificate	Defined award criteria, issue date and verification method.

Curriculum lifecycle

1. Identify learner need and measurable competency.
2. Map prerequisites and place the competency in a learning path.
3. Write objectives using observable actions.
4. Create material with authoritative sources and examples.
5. Design a safe exercise directly supporting the objective.
6. Create assessment, answer key, rubric and feedback.

7 Perform technical, pedagogical, accessibility and safety review.

8 Publish a version with owner and change history; measure outcomes and revise.

The Academy may request a lab through a constrained service, but should not directly control vulnerable machines. The Cyber Range enforces quotas, access policy, time limits and teardown.

7. Cyber Range, CVE labs and CTF

The Cyber Range is the controlled practice environment for container, virtual-machine and hybrid labs, defensive exercises, forensics, intentionally vulnerable applications, CVE reproductions and CTFs. These formats should share one repeatable lifecycle.

Control plane	Lab plane
Identity, curriculum, catalogue, approval and provisioning requests.	Disposable containers, VMs, virtual networks and challenge services.
Stores specs, approved images, quotas, lifecycle state and audit events.	Runs potentially vulnerable workloads and collects scoped exercise results.
Uses protected APIs and approved identity.	Is isolated from production and other learners by default.
Authorises start/stop/reset and records outcomes.	Enforces lab boundaries, resource limits, timeouts and destruction.

End-to-end lab lifecycle

1. Curriculum requirement identifies competency or technique.
2. Choose a safe scenario/CVE and define the intended lesson.
3. Write lab specification: prerequisites, environment, task, success condition.
4. Build a versioned image/template with documented dependencies.
5. Configure infrastructure and network isolation.
6. Review escape paths, secrets, exposed ports, egress and resource exhaustion.
7. Define flags, validation, scoring, hints and evidence collection.
8. QA tests start, exercise path, failure modes, reset and teardown.
9. Approve and publish a versioned catalogue entry.
10. Provision an isolated learner instance and scoped identity.
11. Validate submission, record progress and enforce timeout/reset/destroy.
12. Monitor failures/capacity, maintain versions and retire stale or unsafe labs.

Required lab specification

- Lab ID, title, area, level, type, status and owner.
- Curriculum mapping, objective, prerequisites and technique/CVE references.
- Image/template version, dependencies, resource requirements and network diagram.
- Student task, expected result, validation, flags, scoring, hints and time limit.
- Logging/privacy notes, reset/destroy procedure and failure recovery.
- Security review, QA evidence, release version and retirement plan.

Non-negotiable safety controls

- No direct route from vulnerable labs to production databases, mail systems, admin panels or private networks.
- Default-deny egress; allow only explicitly justified outbound paths.
- Per-student/session identity, quotas, timeouts and cleanup.
- Private management interfaces and authenticated, validated provisioning APIs.
- Student-to-student isolation and separation of management from exercise traffic.
- Tested reset/destroy, monitoring, capacity limits and recovery.
- Back up critical control-plane configuration off-host; avoid treating disposable vulnerable instances as durable assets.

8. Research lifecycle and reproducibility

Research produces defensible knowledge, not just documents. Each project needs a question, method, evidence, interpretation, limitations and review trail. A well-run study can conclude that a hypothesis was not supported.

Research workflow

1. Identify a problem, audience and public benefit.
2. Write a focused research question and explain its relevance.
3. Review prior work and cite primary or authoritative sources.
4. Define methodology, scope, assumptions, metrics and exclusion criteria.
5. Complete ethics, legal, privacy and security review where required.
6. Use public/authorised data or generate synthetic data.
7. Version code, configurations, data schemas and analysis steps.
8. Run experiments and preserve relevant logs, results and provenance.
9. Analyse findings, uncertainty, alternative explanations and limitations.
10. Obtain independent review of method, evidence and reproducibility.
11. Prepare report/publication and complete disclosure/privacy checks.
12. Publish, share privately, revise or withhold based on approval and risk.
13. Archive appropriate artifacts and define retention.

Research project record

Record area	What it contains
Identity	Project ID, title, lead, contributors, supervisor and status.
Question/value	Question, intended benefit, audience and related work.
Method	Design, assumptions, data sources, metrics and procedures.
Ethics/authority	Permission, consent, data classification, restrictions and review decision.
Artifacts	Repository, code version, dependencies, dataset description and hashes if useful.
Evidence/findings	Results, source links, charts, interpretation and limitations.
Review/release	Comments, corrections, approval, publication location and version.
Lifecycle	Follow-up actions, retention, archive and retirement decision.

Research boundaries

- Do not test third-party systems without explicit written authorisation and defined scope.
- Do not use real counselling records, private submissions, production credentials or confidential user data in student projects.
- Use synthetic examples where they answer the question without exposing people.
- Distinguish verified findings from assumptions or speculation.
- For vulnerability research, define responsible disclosure, embargo and publication conditions before release.

9. Innovation and engineering lifecycle

Innovation converts validated findings or needs into useful tools, prototypes or services. Not every research result should become a product; transition requires evidence of value, feasibility, safety and maintenance capacity.

1. Identify a validated problem and intended user.
2. Write a brief with scope, non-goals, requirements and acceptance criteria.
3. Define architecture, data flows, dependencies and threat model.
4. Create a repository with README, ownership, issues and contribution rules.

5 Implement in small changes with code review and automated tests.

6 Run security, privacy, accessibility and operational checks as relevant.

7 Deploy to a private test environment before public release.

8 Document installation, configuration, monitoring, rollback and support.

9 Obtain release approval, publish a version and monitor adoption/defects.

10 Maintain, hand over, archive or retire with a documented decision.

Suggested repository contents: README.md, docs/, src/ or app/, tests/, examples/, licence (when authorised), changelog and contribution guidance. Never commit secrets, personal data, private keys, production dumps or unapproved vulnerable images.

10. Digital platform and data architecture

The digital platform is the coordination and control plane. The public website explains RedKross and serves approved public information. Private dashboards support staff and participants. The lab plane remains separately isolated.

Component	Responsibilities and controls
Public website	Mission, contact, policies, publications and approved resources; secure release, accessibility and minimal collection.
Identity/access	Authentication, roles, session controls, MFA where supported, offboarding and access review.
Programme management	Cohorts, applications, supervisors, assignments and milestones; restrict personal records.
Academy service	Course catalogue, content versions, assessments and progress; role-scoped access.
Research/project registry	Briefs, methods, repositories, evidence, reviewers and decisions.
Lab orchestration API	Start/stop/reset, quotas and lifecycle; private access, strict validation, rate limits and audit.
Evidence/reporting	Status, QA, risk actions and outcomes; aggregate by default and restrict sensitive records.
Notifications	Approved task/status messages; verified destinations and delivery/error handling.
Data/file storage	Records and artifacts; access controls, encryption, retention, integrity and backups.
Operations/monitoring	Health, errors, resource use and security events; restricted access and incident response.

Conceptual data relationships

Person → Programme Membership → Role/Access Grant → Learning Path → Course/Module → Assessment Attempt → Project → Task/Deliverable → Repository/Artifact → Risk Assessment → Review/Approval → Release/Publication → Outcome/Impact Record. A lab instance links to a lab specification, learner session, validation result and teardown record.

Integration rules

- Use stable IDs and explicit relationships, not display names as unique identifiers.
- Define the authoritative system for each record and avoid uncontrolled duplicate spreadsheets.
- Use authenticated APIs with validation, rate limits, error handling and auditable outcomes.
- Separate notification delivery from transaction success; handle retries and partial failures.
- Do not place sensitive data in URLs, general logs, public repositories or analytics events.
- Separate development, test and production and document release/rollback.

11. Security, privacy and safeguarding

Area	Baseline practice
Identity	Unique accounts, least privilege, strong authentication, timely offboarding and periodic access review.
Secrets	No secrets in source control; use restricted secret storage and rotate after suspected exposure.
Network	Separate public, admin, application, mail and lab functions as appropriate; restrict traffic paths.
Application	Input validation, secure sessions, dependency maintenance, rate limiting and security tests.
Data	Classify, minimise, restrict, encrypt appropriately, define retention and securely dispose.
Logging	Capture useful operational/security events without full sensitive payloads.
Backups	Off-host backups of critical records/configuration; restricted access and restore tests.
Change management	Reviewed changes, tested builds, approvals, versioned releases and rollback.
Incident response	Triage, containment, evidence preservation, communication decisions, recovery and lessons.
Safeguarding	Conduct rules, reporting routes, respectful communication and suitable supervision.

12. Quality assurance and release

Each deliverable needs acceptance criteria before work starts, evidence that the criteria were tested, a reviewer and a clear decision. QA must be proportionate to risk, but should never be an afterthought.

Gate	Questions
Scope	Is it defined, relevant, bounded and owned?
Technical correctness	Do claims, code, instructions and examples work as stated?
Evidence/reproducibility	Can a reviewer trace sources and repeat key steps?
Security/privacy	Are access, data, secrets, exposure and misuse risks addressed?
Ethics/legal/safeguarding	Is the activity authorised and appropriate?
Accessibility/usability	Can the intended audience use the content or service?
Operations	Are monitoring, support, rollback, reset or retirement defined?
Release	Is the approval recorded and version identified?

Publication paths

- **Public:** after content, privacy, legal, security and QA checks appropriate to risk.
- **Internal:** restricted to the intended programme/reviewers and marked draft where applicable.
- **Responsible disclosure:** defined contact, scope, embargo, coordination and publication process.
- **Hold/reject:** if evidence, authority, ownership or acceptable risk is missing.

A release record should include version, date, owner, reviewer, tests, known limitations, approval decision, release location, rollback/retirement method and follow-up actions.

13. Operating cadence and end-to-end workflows

Cadence	Activity	Evidence
At start	Confirm task, owner, scope, dependencies, acceptance criteria and risk class.	Task/project record.
Daily/as agreed	Brief update: done, next, blocker and evidence link.	Status/blocker trail.
Weekly	Review deliverables, sources, tests, risks and milestones.	Review notes and updated plan.
Milestone	Reviewer checks acceptance criteria and returns approve/revise/reject.	QA record and actions.
Before release	Complete risk-appropriate security/privacy/legal/ethics checks.	Release checklist and approval.
Programme cycle	Review capacity, incidents, outcomes and feedback.	Programme health report.
Completion	Handover, access review, retention, feedback and lessons.	Closure checklist.

Workflow A — student to research contribution

Onboard → assign learning path → complete prerequisites → practise in approved lab → define scoped question → approve method/data → create repository → run experiment → peer review → revise → publish/archive → record competency and impact.

Workflow B — CVE/technique to lab

Curriculum need → safe scenario → specification → image build → network isolation → security review → assessment/validation → QA → release → provisioning → learner practice → scoring → reset/destroy → monitoring → version or retire.

Workflow C — research to useful software

Validated finding → evidence review → user need → engineering brief → threat model → repository/tests → prototype → QA → privacy/security review → approved deployment → adoption monitoring → maintain or retire.

Workflow D — incident

Report → triage → classify → contain → preserve evidence → notify owners → recover safely → decide communication → root cause → corrective actions → verify closure → update controls.

14. Deliverables and documentation standards

Artefact	Minimum contents
Programme charter	Purpose, audience, scope, roles, resources, outcomes, risks and decision owner.
Project brief	Problem, objective, non-goals, owner, stakeholders, deliverables, timeline and acceptance criteria.
Risk assessment	Assets, threats, likelihood/impact, mitigation, residual risk, approver and review date.
Lab specification	Curriculum mapping, environment, network, task, validation, scoring, reset, logging, QA and version.
Research protocol	Question, related work, method, data, metrics, ethics/authority, limitations and reproduction.
Technical report	Summary, problem, method, evidence, results, limitations, recommendations and references.
Repository README	Purpose, status, setup, configuration, usage, tests, expected output and limitations.
QA record	Acceptance criteria, tests, evidence, defects, reviewer, outcome and re-test status.
Release record	Version, date, approver, changes, issues, deployment/rollback and support.
Incident record	Timeline, impact, containment, evidence, decisions, cause and corrective actions.
Impact report	Baseline, activities, outputs, outcomes, measurement method and limitations.

Student repository requirements

- Use a dedicated repository with clear name, description and README.
- Keep commits meaningful and maintain reproducible instructions.
- Include tests or a validation checklist appropriate to the work.
- Document dependencies, sample inputs and expected results.
- Do not publish private student details, credentials, confidential datasets or unapproved lab artifacts.
- Use a licence only when the owner has authority and the intended distribution is clear.
- Share the repository with the designated supervisor/research contacts and state whether it is private or public.

15. Implementation roadmap

Phase	Focus	Exit criteria
0 — Baseline	Confirm mission, owners, current assets, policies, risk taxonomy and resource constraints.	Approved principles, backlog and baseline.
1 — Governance	Project charters, onboarding/offboarding, risk records, QA, incident and retention processes.	Work can start and close with consistent records.
2 — Control plane	Secure identity, project registry, task/evidence tracking, audit and reporting.	Least privilege, tested backup/restore and rollback.
3 — Academy	Taxonomy, templates, initial paths, assessments and content review.	Initial learning paths have reviewed objectives and exercises.
4 — Cyber Range	Container lab, VM lab and multi-machine/hybrid prototype where feasible.	Isolation, provisioning, validation, reset/destroy and failure tests pass.
5 — Research/engineering	Reproducible templates, research review, publication and prototype workflow.	One end-to-end project has reproducible evidence and approval.
6 — Pilot	Controlled cohort, feedback, operational measurements and fixes.	Pilot review and risk actions resolved or accepted.
7 — Scale	Expand labs/courses/projects and automate repeatable operations.	Capacity, ownership, support, cost and safety stay within limits.

Build in dependency order. Governance, ownership, security and documentation are prerequisites for scaling labs, courses, projects or users. A successful demo is not enough: require repeatable setup, cleanup, monitoring, recovery and support.

16. Metrics, risks and open decisions

Metrics should inform decisions, not reward activity at the expense of quality. Establish baselines before setting numerical targets.

Risk	Potential impact	Mitigation
Unclear ownership	Stalled work and inconsistent approvals.	One accountable owner per project/decision and a backup.
Unsafe lab isolation	Production exposure or cross-student compromise.	Separate lab plane, default-deny network, isolation tests and teardown.
Unclear authorisation	Unauthorised testing and legal/reputational harm.	Written scope and permission; stop if unclear.
Sensitive data leakage	Privacy harm and loss of trust.	Minimisation, synthetic data, redaction, restricted access and retention.
Unreviewed publication	Incorrect claims or disclosure harm.	Risk-based review and recorded approval.
Overloaded supervisors	Poor feedback and missed deadlines.	Capacity planning, work-in-progress limits and reviewer allocation.
Fragile automation	Lost/duplicated records or incorrect status.	Tests, error handling, monitoring and recovery procedures.
Unclear retention	Excess data or premature loss.	Approve schedules by record type before automating deletion.
Scaling too early	Downtime, cost and support failures.	Stage gates, capacity tests and operational readiness.

Dashboard indicators

- Workload: active projects, overdue milestones, blockers and reviewer capacity.
- Quality: QA pass rate, recurring defects, rework and time to close findings.
- Security: high-risk actions, access review, incidents, restore tests and isolation failures.
- Academy: progression, assessment outcomes, exercise completion and accessibility defects.
- Cyber Range: provisioning success, start time, teardown success, resource use and capacity.
- Research/innovation: reproducible projects, reviewed outputs, releases, adoption and maintenance ownership.
- Impact: reach, reuse, partner feedback, demonstrated skill gains and community benefit.

Open decisions requiring formal approval

- Mission statement, beneficiaries and measurable outcomes.
- Named accountable owners for each domain and deputy arrangements.
- Risk taxonomy, approval authority, escalation thresholds and active R0/R1+ QA policy.
- Privacy notice, data inventory, retention schedule, deletion and incident responsibilities.
- Safeguarding, conduct process, reporting routes and supervision requirements.
- Cyber Range isolation, egress, quotas, runtime and cleanup guarantees.
- Identity, MFA, admin access, access reviews and production change controls.
- Repository/publication licences, intellectual property and contribution terms.
- Research ethics, responsible disclosure and publication approval.
- Budget, infrastructure capacity, support, backup objectives and recovery targets.
- Maintenance and retirement owners for courses, labs, repositories and deployed tools.

Final architecture statement

Govern → Onboard → Learn → Practise → Investigate → Build → Assure → Release/Share → Measure → Improve. Every activity has an owner and scope; higher-risk work receives stronger review; cyber environments are isolated; research and software are reproducible; and outputs are judged by quality, safety and public benefit—not volume alone.

End of blueprint · Working architecture reference · Review and approve before treating proposed controls as formal policy.